

Let R be a Commutative Ring with $1 \neq 0$, in this paper.

Def. In R , define a set

$$\eta(R) \stackrel{\text{def}}{=} \{x \in R \mid x \text{ is nilpotent}\}. \quad (\text{i.e., } x^m = 0 \text{ for some } m \in \mathbb{N})$$

Actually $\eta(R)$ is an ideal.

Given $a, b \in \eta(R)$, $a^m = b^n = 0$ for some $m, n \in \mathbb{N}$, so

$$(a+b)^{m+n} = \sum_{k=0}^{m+n} \binom{m+n}{k} \cdot a^{m+n-k} \cdot b^k$$

If $0 \leq k < n$, then $m+n-k > m$, so $a^{m+n-k} = 0$

If $k \geq n$, then $b^k = 0$. Therefore, $(a+b)^{m+n} = 0$, or $a+b \in \eta(R)$.

And, $(-a)^m = (-1)^m \cdot a^m = 0$, so $-a \in \eta(R)$.

Also, since $(ab)^{mn} = a^{mn} \cdot b^{mn} = 0$ by the Commutativity, $ab \in \eta(R)$. Subring.

Given $r \in R$ and $a \in \eta(R)$ with $a^m = 0$, $(ra)^m = r^m a^m = r^m \cdot 0 = 0$, so $ra \in \eta(R)$. Ideal.

Lemma. $R/\eta(R)$ has no nilpotent element except zero.

Proof. Let $a + \eta(R) \in R/\eta(R)$ be given. If $(a + \eta(R))^n = 0$ for some $n \in \mathbb{N}$, then

$$(a + \eta(R))^n = a^n + \eta(R) = 0 + \eta(R) \Leftrightarrow a^n \in \eta(R)$$

Then, a^n is a nilpotent element so a is nilpotent, or $a \in \eta(R)$.

Thus, the only nilpotent element is $0 + \eta(R)$ in $R/\eta(R)$.

Prop. Let R be a Commutative Ring with $1 \neq 0$. Given $p(x) = a_n x^n + \dots + a_1 x + a_0 \in R[x]$,

$p(x)$ is a unit in $R[x]$ iff a_0 is unit and $a_1, \dots, a_n$ are nilpotent.

$p(x)$ is a nilpotent in $R[x]$ iff $a_0, \dots, a_n$ are nilpotent.

Proof. Since sum of nilpotent, if $a_1, \dots, a_n$ are nilpotent, then $n = a_1 x + \dots + a_n x^n$ is nilpotent.

So, $a_0 + n$ is a unit, being sum of unit and nilpotent.

Conversely, if $p(x)$ is a unit, then for some $q(x) = b_m x^m + \dots + b_0 \in R[x]$, $p(x) \cdot q(x) = 1$.

So, $a_n b_m x^{n+m} = 0$,

Thm. Let R be a Commut. Ring with $1 \neq 0$.

Then, $\mathcal{N}(R) \subseteq \bigcap_{\substack{P \subset R \\ P \text{ is prime ideal}}} P$. (Actually, these are same).

Proof. Let P be a Prime Ideal. Suppose that for some non-zero nilpotent $a \in R$, $a \notin P$. That is, $a+P \neq P$ in R/P . Since a is nilpotent, $m \in \mathbb{N}$ be the smallest number s.t. $a^m = 0$. Now, $P = a^m + P = (a+P) \cdot (a^{m-1} + P)$,

by the minimality of m , $a^{m-1} + P \neq P$, so $a+P$ is zero divisor.

But, since P is prime ideal, R/P is integral domain. So Contradiction.

Another proof: $0 = a^m = a \cdot a^{m-1} \in P$, so a or $a^{m-1} \in P$.

$a \in P$, then clear. $a^{m-1} \in P$, then a or $a^{m-2} \in P$.

Inductively, $a \in P$.

Prop. If $a \in R$ is nilpotent, then $1-ab$ is a unit, for any $b \in R$.

Proof. Let $m \in \mathbb{N}$ s.t. $a^m = 0$. Then,

$$(1-ab) \cdot (1+ab-(ab)^2+\dots+(-ab)^{m-1}) = 1 + (-1)^m (ab)^m = 1 + 0 = 1.$$

Def. Let $I \subseteq R$ be an Ideal. Define the radical of I

$$\text{rad } I \stackrel{\text{def}}{=} \{r \in R \mid r^n \in I \text{ for some } n \in \mathbb{N}\}.$$

Then, $\text{rad } I$ is an Ideal of R satisfying

$$(\text{rad } I)/I = \eta(R/I).$$

And, I is called a radical Ideal if $\text{rad } I = I$.

Proof. 1) $\text{rad } I$ is an Ideal.

Given $a, b \in \text{rad } I$, $a^m, b^n \in I$ for some $m, n \in \mathbb{N}$. So,

$$(a+b)^{m+n} = \sum_{k=0}^{m+n} \binom{m+n}{k} a^{m+n-k} \cdot b^k \in I,$$

because since I is ideal, if $0 \leq k < n$, $\binom{m+n}{k} \cdot a^{m+n-k} \cdot b^k = \binom{m+n}{k} \cdot b^k \cdot a^{n-k} \cdot a^m \in I$,
and if $k \geq n$, $\binom{m+n}{k} \cdot a^{m+n-k} \cdot b^k = \binom{m+n}{k} \cdot a^{m+n-k} \cdot b^n \in I$. So, the sum is contained in I .

And, similarly, $(ab)^{m+n} = a^{m+n} \cdot b^{m+n} \in I$. $\hookrightarrow$ Subring.

And, given $r \in R$ and $a \in \text{rad } I$,

$$(ra)^m = r^m a^m \in I \quad \downarrow \text{Ideal.}$$

Clearly, $I \subseteq \text{rad } I$. Define a map

$$\ell: \text{rad } I \rightarrow \eta(R/I): a \mapsto a+I$$

Then, ℓ is well-defined onto Homomorphism, because

given $a \in \text{rad } I$, $a^m \in I$ for some $m \in \mathbb{N}$, so $(a+I)^m = a^m + I = I$, or $a+I \in \eta(R/I)$.

So well-defined. Homomorphism is clear, and onto is clear. Moreover,

$$\ell(a) = 0 \Leftrightarrow a+I = I \Leftrightarrow a \in I, \text{ so } \ker \ell = I.$$

Finally, the first-Isomorphism gives $\text{rad } I/I \cong \eta(R/I)$.

Actually, $\bar{\ell}: \text{rad } I/I \rightarrow \eta(R/I): a+I \mapsto a+I$

is identifying, $\text{rad } I/I = \eta(R/I)$.

Nilpotent with the Polynomial Ring.

Lemma. Let R be a Ring with identity, and $a, b \in R$.

If a is nilpotent and b is unit and a, b commute, then $a+b$ is unit.

Proof. Choose $n \in \mathbb{N}$ s.t. $a^n = 0$. Then, since $a = abb^{-1} = bab^{-1} \Rightarrow b^{-1}a = ab^{-1}$, and $b^n - a^n = (b+a) \cdot \left(\sum_{k=0}^{n-1} (-1)^k b^{n-k-1} a^k \right) = b^n$, therefore

$$(a+b)^{-1} = \left(\sum_{k=0}^{n-1} (-1)^k b^{n-k-1} a^k \right) \cdot b^{-n}.$$